AERIS Expert Review Packet

Anomaly

Source / metric	tceq / Sulfur dioxide (so2)
Observed / expected baseline	1.8 / -0.402113 ppb
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-07-20 22:00 UTC
Location	29.7337, -95.2576
Severity	severe
Detectors	zscore, stl, isolation_forest

The baseline is a mean of this series' own recent history, so it can fall below zero for a metric the network reports near or below its detection limit.

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	9 / 897	29.01 to 100; mean 72.4599	41.62 at 2026-07-20 21:55Z; 5 min before event
asos	Air temperature (temperature)	degC	9 / 897	22 to 37; mean 29.464	35 at 2026-07-20 21:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	9 / 1163	0 to 360; mean 188.942	110 at 2026-07-20 21:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	9 / 1209	0 to 7.20222; mean 2.4063	2.05778 at 2026-07-20 21:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	10 / 120	5915.17 to 5937.49; mean 5926.11	5923.64 at 2026-07-21 00:00Z; 2 h after event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	10 / 120	17.7446 to 1614.77; mean 603.657	1200.05 at 2026-07-21 00:00Z; 2 h after event
noaa_gfs	Precipitable water (precipitable_water)	mm	10 / 120	29.3968 to 46.5607; mean 35.8788	37.1017 at 2026-07-21 00:00Z; 2 h after event
noaa_gfs	Surface pressure (surface_pressure)	hPa	10 / 120	1005.31 to 1015.78; mean 1010.52	1009.39 at 2026-07-21 00:00Z; 2 h after event
noaa_gfs	850-hPa temperature (t_850)	degC	10 / 120	19.7215 to 23.3589; mean 22.1272	21.282 at 2026-07-21 00:00Z; 2 h after event
noaa_gfs	10-m eastward wind (u_10m)	m/s	10 / 120	-2.78129 to 3.37582; mean 1.3835	0.318389 at 2026-07-21 00:00Z; 2 h after event
noaa_gfs	10-m northward wind (v_10m)	m/s	10 / 120	-1.86296 to 4.7729; mean 1.2267	2.58333 at 2026-07-21 00:00Z; 2 h after event

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	Ozone (ozone)	ppm	17 / 1106	-0.004 to 0.094; mean 0.0207	0.057 at 2026-07-20 22:00Z; at event time
openaq	PM10 (pm10)	ug/m3	2 / 142	29 to 68; mean 44.4789	41 at 2026-07-20 22:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	77 / 2969	0.3 to 28.8; mean 7.2232	13.4 at 2026-07-20 22:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 357	0 to 100; mean 20.5826	6 at 2026-07-20 22:01Z; 2 min after event
openweather	Relative humidity (humidity)	percent	5 / 357	40 to 96; mean 72.7255	52 at 2026-07-20 22:01Z; 2 min after event
openweather	Precipitation rate (precipitation)	mm/h	5 / 357	0 to 0; mean 0	0 at 2026-07-20 22:01Z; 2 min after event
openweather	Surface pressure (pressure)	hPa	5 / 357	1009 to 1017; mean 1012.92	1012 at 2026-07-20 22:01Z; 2 min after event
openweather	Air temperature (temperature)	degC	5 / 357	24.14 to 37.79; mean 30.5353	36.3 at 2026-07-20 22:01Z; 2 min after event
openweather	Wind direction (wind_direction)	degree	5 / 357	0 to 343; mean 212.213	155 at 2026-07-20 22:01Z; 2 min after event
openweather	Wind speed (wind_speed)	m/s	5 / 357	0.4 to 5.9; mean 1.9989	2.86 at 2026-07-20 22:01Z; 2 min after event
purpleair	PM2.5 (pm25)	ug/m3	72 / 5159	0 to 149.3; mean 9.0727	7.7 at 2026-07-20 21:59Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-20 19:49Z; 2 h 10 min before event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-20 19:49Z; 2 h 10 min before event
sentinel5p	CO column (s5p_co_column)	mol/m^2	6 / 6	0.0254212 to 0.0265722; mean 0.0261	0.0262849 at 2026-07-20 19:49Z; 2 h 10 min before event
sentinel5p	CO granules available (s5p_co_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-20 19:49Z; 2 h 10 min before event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	5 / 5	0.000176599 to 0.000224886; mean 0.0002	0.000213688 at 2026-07-20 19:49Z; 2 h 10 min before event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-07-20 19:49Z; 2 h 10 min before event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	3 / 3	3.22877e-05 to 5.87309e-05; mean 0	5.83485e-05 at 2026-07-20 19:49Z; 2 h 10 min before event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	3 / 3	1 to 1; mean 1	1 at 2026-07-20 19:49Z; 2 h 10 min before event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-20 19:49Z; 2 h 10 min before event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	5 / 5	-8.60324e-06 to 4.32022e-05; mean 0	-8.60324e-06 at 2026-07-20 19:49Z; 2 h 10 min before event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-07-20 19:49Z; 2 h 10 min before event
tceq	Carbon monoxide (co)	ppm	3 / 185	0 to 0.7; mean 0.1762	0.2 at 2026-07-20 22:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Nitrogen dioxide (no2)	ppb	12 / 701	-2.5 to 44.3; mean 5.175	5.8 at 2026-07-20 22:00Z; at event time
tceq	Sulfur dioxide (so2)	ppb	3 / 182	-0.6 to 6.9; mean 0.1143	1.8 at 2026-07-20 22:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	07-19 06Z 95.98, 12Z 75.18, 18Z 54.34, 07-20 00Z 79.67, 06Z 94.44, 12Z 77.08, 18Z 48.96, 07-21 00Z 76.41, 06Z 93.27, 12Z 74.93, 18Z 40.31, 07-22 00Z 64.01, 06Z 78.46
asos	Air temperature (temperature)	07-19 06Z 24.33, 12Z 29.45, 18Z 33.97, 07-20 00Z 28.69, 06Z 24.95, 12Z 28.8, 18Z 34.38, 07-21 00Z 29.01, 06Z 25.08, 12Z 28.86, 18Z 35.33, 07-22 00Z 30.11, 06Z 27.1
asos	Wind direction (wind_direction)	07-19 06Z 44.24, 12Z 186.7, 18Z 139.9, 07-20 00Z 167.3, 06Z 133.3, 12Z 225.3, 18Z 176.1, 07-21 00Z 180.4, 06Z 201.3, 12Z 259.9, 18Z 231.5, 07-22 00Z 196.5, 06Z 237.5
asos	Wind speed (wind_speed)	07-19 06Z 0.5296, 12Z 2.361, 18Z 2.239, 07-20 00Z 2.895, 06Z 1.554, 12Z 2.562, 18Z 2.234, 07-21 00Z 2.832, 06Z 2.166, 12Z 3.44, 18Z 2.344, 07-22 00Z 2.186, 06Z 2.826
noaa_gfs	500-hPa geopotential height (gh_500)	07-19 12Z 5923, 18Z 5926, 07-20 00Z 5922, 06Z 5930, 12Z 5920, 18Z 5928, 07-21 00Z 5924, 06Z 5926, 12Z 5917, 18Z 5932, 07-22 00Z 5930, 06Z 5935
noaa_gfs	Planetary boundary layer height (pbl_height)	07-19 12Z 166.1, 18Z 1342, 07-20 00Z 724.3, 06Z 227.6, 12Z 139.9, 18Z 1265, 07-21 00Z 780.6, 06Z 208.9, 12Z 207.2, 18Z 1217, 07-22 00Z 849.1, 06Z 116.5
noaa_gfs	Precipitable water (precipitable_water)	07-19 12Z 34.74, 18Z 36.22, 07-20 00Z 39.11, 06Z 33.86, 12Z 30.15, 18Z 32.95, 07-21 00Z 36.06, 06Z 33.96, 12Z 32, 18Z 35.81, 07-22 00Z 40.3, 06Z 45.38
noaa_gfs	Surface pressure (surface_pressure)	07-19 12Z 1014, 18Z 1013, 07-20 00Z 1010, 06Z 1012, 12Z 1012, 18Z 1012, 07-21 00Z 1009, 06Z 1009, 12Z 1010, 18Z 1010, 07-22 00Z 1007, 06Z 1008
noaa_gfs	850-hPa temperature (t_850)	07-19 12Z 21.95, 18Z 21.53, 07-20 00Z 20.92, 06Z 22.7, 12Z 22.64, 18Z 22.34, 07-21 00Z 21.4, 06Z 22.98, 12Z 22.39, 18Z 22.32, 07-22 00Z 21.86, 06Z 22.5
noaa_gfs	10-m eastward wind (u_10m)	07-19 12Z 0.7036, 18Z 1.36, 07-20 00Z -0.5413, 06Z 1.477, 12Z 1.61, 18Z 1.903, 07-21 00Z -0.2606, 06Z 1.821, 12Z 2.649, 18Z 2.383, 07-22 00Z 1.138, 06Z 2.358
noaa_gfs	10-m northward wind (v_10m)	07-19 12Z 1.625, 18Z 1.541, 07-20 00Z 2.719, 06Z 2.416, 12Z 0.5624, 18Z 0.6997, 07-21 00Z 2.019, 06Z 2.349, 12Z 0.3206, 18Z -0.6552, 07-22 00Z -0.08996, 06Z 1.214
openaq	Ozone (ozone)	07-19 06Z 0.007353, 12Z 0.01435, 18Z 0.03404, 07-20 00Z 0.01726, 06Z 0.006953, 12Z 0.01077, 18Z 0.04282, 07-21 00Z 0.01878, 06Z 0.006526, 12Z 0.01157, 18Z 0.0444, 07-22 00Z 0.02872, 06Z 0.006833
openaq	PM10 (pm10)	07-19 06Z 34.75, 12Z 35.75, 18Z 46.83, 07-20 00Z 45.25, 06Z 44.75, 12Z 49.67, 18Z 50.75, 07-21 00Z 43.55, 06Z 39.83, 12Z 41.33, 18Z 49.09, 07-22 00Z 46.83, 06Z 43.25
openaq	PM2.5 (pm25)	07-19 06Z 8.934, 12Z 6.381, 18Z 8.076, 07-20 00Z 7.664, 06Z 8.644, 12Z 7.871, 18Z 6.551, 07-21 00Z 6.747, 06Z 7.397, 12Z 6.958, 18Z 5.658, 07-22 00Z 6.102, 06Z 6.149
openweather	Cloud cover (cloud_cover)	07-19 06Z 91.9, 12Z 94.52, 18Z 84.9, 07-20 00Z 12.83, 06Z 5.645, 12Z 3.833, 18Z 2.967, 07-21 00Z 3.69, 06Z 0.3226, 12Z 1.138, 18Z 1.806, 07-22 00Z 3.267, 06Z 13.75
openweather	Relative humidity (humidity)	07-19 06Z 92.9, 12Z 81.63, 18Z 60.1, 07-20 00Z 74.79, 06Z 90.77, 12Z 80.67, 18Z 53.33, 07-21 00Z 71.24, 06Z 89.23, 12Z 79.59, 18Z 48, 07-22 00Z 61.2, 06Z 78
openweather	Precipitation rate (precipitation)	07-19 06Z 0, 12Z 0, 18Z 0, 07-20 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-21 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-22 00Z 0, 06Z 0
openweather	Surface pressure (pressure)	07-19 06Z 1016, 12Z 1017, 18Z 1014, 07-20 00Z 1013, 06Z 1014, 12Z 1015, 18Z 1012, 07-21 00Z 1012, 06Z 1012, 12Z 1013, 18Z 1011, 07-22 00Z 1010, 06Z 1010
openweather	Air temperature (temperature)	07-19 06Z 25.32, 12Z 28.9, 18Z 35.34, 07-20 00Z 30.62, 06Z 25.88, 12Z 28.74, 18Z 35.79, 07-21 00Z 30.97, 06Z 25.89, 12Z 28.79, 18Z 36.46, 07-22 00Z 31.98, 06Z 27.81
openweather	Wind direction (wind_direction)	07-19 06Z 178.5, 12Z 213.6, 18Z 184.9, 07-20 00Z 186.7, 06Z 194.4, 12Z 239.7, 18Z 215.6, 07-21 00Z 183.5, 06Z 198.9, 12Z 249.6, 18Z 259.8, 07-22 00Z 234.3, 06Z 187.7

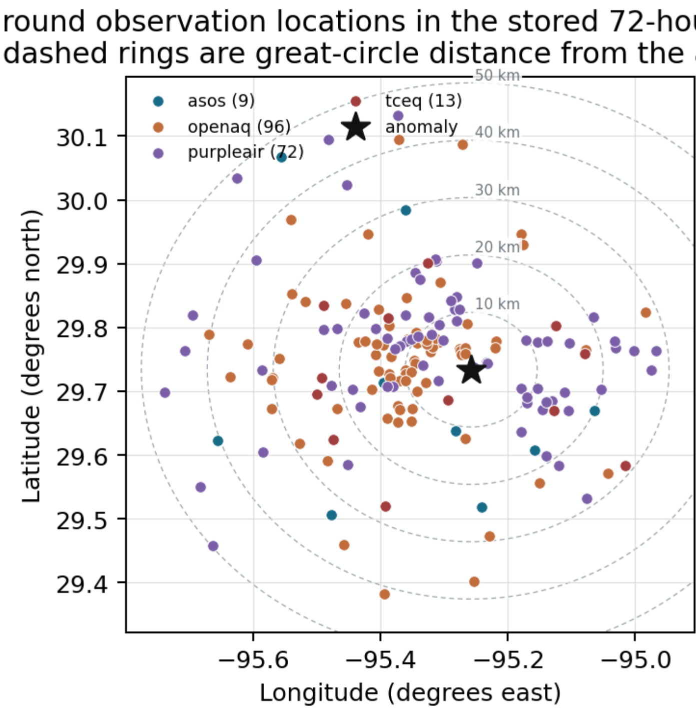
Source	Metric	Values
openweather	Wind speed (wind_speed)	07-19 06Z 1.702, 12Z 1.676, 18Z 1.968, 07-20 00Z 2.389, 06Z 1.895, 12Z 1.845, 18Z 1.766, 07-21 00Z 2.368, 06Z 2.22, 12Z 2.04, 18Z 1.646, 07-22 00Z 2.314, 06Z 1.945
purpleair	PM2.5 (pm25)	07-19 06Z 10.87, 12Z 8.744, 18Z 8.76, 07-20 00Z 9.393, 06Z 10.97, 12Z 9.927, 18Z 7.759, 07-21 00Z 9.001, 06Z 9.814, 12Z 8.67, 18Z 7.789, 07-22 00Z 9.72, 06Z 7.196
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	07-19 18Z 1, 07-20 18Z 1, 07-21 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	07-19 18Z 1, 07-20 18Z 1, 07-21 18Z 1
sentinel5p	CO column (s5p_co_column)	07-19 18Z 0.02551, 07-20 18Z 0.02628, 07-21 18Z 0.02656
sentinel5p	CO granules available (s5p_co_granule_available)	07-19 18Z 1, 07-20 18Z 1, 07-21 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	07-19 18Z 0.0001766, 07-20 18Z 0.000213, 07-21 18Z 0.0002243
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	07-19 18Z 1, 07-20 18Z 1, 07-21 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	07-19 18Z 3.229e-05, 07-20 18Z 5.835e-05, 07-21 18Z 5.873e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	07-19 18Z 1, 07-20 18Z 1, 07-21 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	07-19 18Z 1, 07-20 18Z 1, 07-21 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	07-19 18Z 2.991e-06, 07-20 18Z 1.73e-05, 07-21 18Z 3.63e-05
sentinel5p	SO2 granules available (s5p_so2_granule_available)	07-19 18Z 1, 07-20 18Z 1, 07-21 18Z 1
tceq	Carbon monoxide (co)	07-19 06Z 0.2167, 12Z 0.2111, 18Z 0.2, 07-20 00Z 0.2667, 06Z 0.2056, 12Z 0.2125, 18Z 0.1722, 07-21 00Z 0.2222, 06Z 0.1222, 12Z 0.1556, 18Z 0.1056, 07-22 00Z 0.21, 06Z 0.1
tceq	Nitrogen dioxide (no2)	07-19 06Z 3.453, 12Z 2.541, 18Z 3.297, 07-20 00Z 3.545, 06Z 4.577, 12Z 6.21, 18Z 6.136, 07-21 00Z 6.723, 06Z 5.171, 12Z 5.402, 18Z 4.761, 07-22 00Z 10.57, 06Z 6.231
tceq	Sulfur dioxide (so2)	07-19 06Z -0.575, 12Z 0.1056, 18Z 0.4389, 07-20 00Z -0.2875, 06Z -0.3471, 12Z -0.175, 18Z 1.906, 07-21 00Z 1.213, 06Z -0.2778, 12Z -0.275, 18Z -0.1611, 07-22 00Z -0.1222, 06Z -0.2933

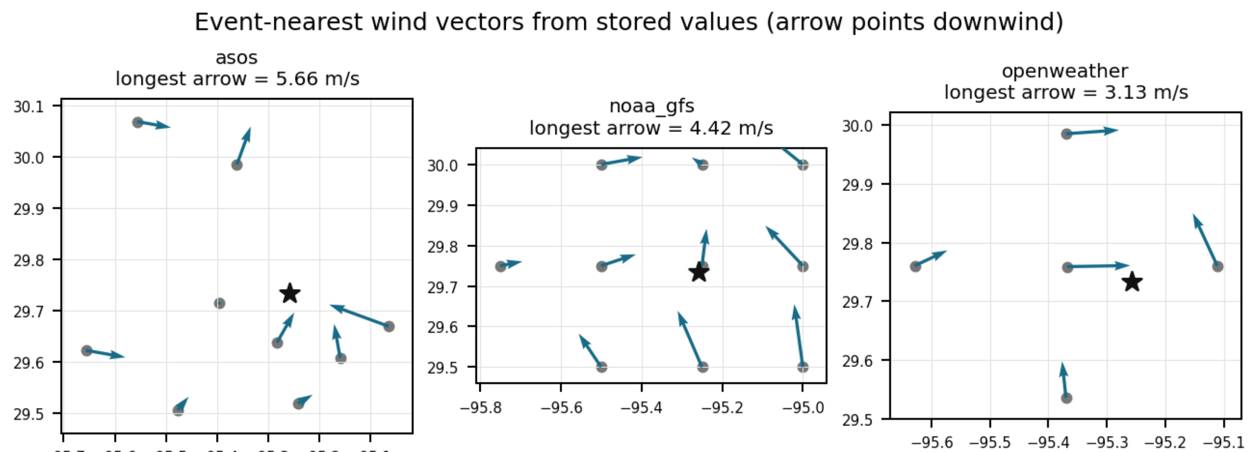
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	Relative humidity (humidity)	HOU (10.965 km) 71.06, MCJ (13.448 km) 67.54, EFD (16.998 km) 69.85, T41 (20.02 km) 72.7, LVJ (23.937 km) 73.28, IAH (29.593 km) 70, AXH (33.015 km) 72.15, SGR (40.48 km) 79.61, +1 more
asos	Air temperature (temperature)	HOU (10.965 km) 29.87, MCJ (13.448 km) 30.27, EFD (16.998 km) 30.29, T41 (20.02 km) 29.75, LVJ (23.937 km) 29.27, IAH (29.593 km) 30.07, AXH (33.015 km) 29.11, SGR (40.48 km) 28.39, +1 more
asos	Wind direction (wind_direction)	HOU (10.965 km) 223.1, MCJ (13.448 km) 211.2, EFD (16.998 km) 197.9, T41 (20.02 km) 203, LVJ (23.937 km) 169.3, IAH (29.593 km) 191.5, AXH (33.015 km) 128.7, SGR (40.48 km) 201.5, +1 more
asos	Wind speed (wind_speed)	HOU (10.965 km) 3.197, MCJ (13.448 km) 3.108, EFD (16.998 km) 3.394, T41 (20.02 km) 3.087, LVJ (23.937 km) 1.772, IAH (29.593 km) 2.279, AXH (33.015 km) 1.051, SGR (40.48 km) 2.504, +1 more
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.25 (1.952 km) 5926, gfs:29.75,-95.50 (23.473 km) 5927, gfs:29.75,-95.00 (24.937 km) 5925, gfs:29.50,-95.25 (26.0 km) 5926, gfs:30.00,-95.25 (29.617 km) 5926, gfs:29.50,-95.50 (34.993 km) 5926, gfs:29.50,-95.00 (35.994 km) 5925, gfs:30.00,-95.50 (37.722 km) 5927, +2 more

Source	Metric	Values
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.25 (1.952 km) 790.8, gfs:29.75,-95.50 (23.473 km) 854.4, gfs:29.75,-95.00 (24.937 km) 564.3, gfs:29.50,-95.25 (26.0 km) 429.8, gfs:30.00,-95.25 (29.617 km) 778.2, gfs:29.50,-95.50 (34.993 km) 438.8, gfs:29.50,-95.00 (35.994 km) 478.2, gfs:30.00,-95.50 (37.722 km) 785, +2 more
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.25 (1.952 km) 36.08, gfs:29.75,-95.50 (23.473 km) 35.04, gfs:29.75,-95.00 (24.937 km) 36.89, gfs:29.50,-95.25 (26.0 km) 35.48, gfs:30.00,-95.25 (29.617 km) 36.71, gfs:29.50,-95.50 (34.993 km) 34.74, gfs:29.50,-95.00 (35.994 km) 36.36, gfs:30.00,-95.50 (37.722 km) 35.99, +2 more
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.25 (1.952 km) 1011, gfs:29.75,-95.50 (23.473 km) 1010, gfs:29.75,-95.00 (24.937 km) 1012, gfs:29.50,-95.25 (26.0 km) 1012, gfs:30.00,-95.25 (29.617 km) 1010, gfs:29.50,-95.50 (34.993 km) 1011, gfs:29.50,-95.00 (35.994 km) 1013, gfs:30.00,-95.50 (37.722 km) 1008, +2 more
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.25 (1.952 km) 22.03, gfs:29.75,-95.50 (23.473 km) 22.05, gfs:29.75,-95.00 (24.937 km) 22.05, gfs:29.50,-95.25 (26.0 km) 22.26, gfs:30.00,-95.25 (29.617 km) 21.88, gfs:29.50,-95.50 (34.993 km) 22.25, gfs:29.50,-95.00 (35.994 km) 22.11, gfs:30.00,-95.50 (37.722 km) 22.09, +2 more
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.25 (1.952 km) 1.371, gfs:29.75,-95.50 (23.473 km) 1.818, gfs:29.75,-95.00 (24.937 km) 0.678, gfs:29.50,-95.25 (26.0 km) 1.478, gfs:30.00,-95.25 (29.617 km) 0.9855, gfs:29.50,-95.50 (34.993 km) 1.481, gfs:29.50,-95.00 (35.994 km) 1.516, gfs:30.00,-95.50 (37.722 km) 2.054, +2 more
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.25 (1.952 km) 1.315, gfs:29.75,-95.50 (23.473 km) 1.054, gfs:29.75,-95.00 (24.937 km) 1.263, gfs:29.50,-95.25 (26.0 km) 1.645, gfs:30.00,-95.25 (29.617 km) 0.6752, gfs:29.50,-95.50 (34.993 km) 1.566, gfs:29.50,-95.00 (35.994 km) 1.8, gfs:30.00,-95.50 (37.722 km) 0.7177, +2 more
openaq	Ozone (ozone)	3378 (0.021 km) 0.01843, 2039 (5.231 km) 0.0231, 325 (6.369 km) 0.0062, 2046 (10.803 km) 0.01976, 320 (12.059 km) 0.01807, 1319333 (14.033 km) 0.02407, 332 (14.34 km) 0.02468, 3214 (14.86 km) 0.02547, +9 more
openaq	PM10 (pm10)	15852419 (7.017 km) 37.46, 271 (14.34 km) 51.49
openaq	PM2.5 (pm25)	15539682 (0.005 km) 9.665, 8967123 (0.058 km) 10.97, 8967479 (0.058 km) 10.81, 9489522 (2.89 km) 8.424, 8967065 (2.933 km) 9.064, 12178865 (3.713 km) 9.522, 9201803 (3.931 km) 9.858, 10252270 (4.015 km) 10.13, +69 more
openweather	Cloud cover (cloud_cover)	grid:center (10.991 km) 18.32, grid:east (14.451 km) 20.38, grid:south (24.535 km) 20.92, grid:north (29.962 km) 20.24, grid:west (35.93 km) 23.01
openweather	Relative humidity (humidity)	grid:center (10.991 km) 70.66, grid:east (14.451 km) 77.07, grid:south (24.535 km) 72.62, grid:north (29.962 km) 70.99, grid:west (35.93 km) 72.29
openweather	Precipitation rate (precipitation)	grid:center (10.991 km) 0, grid:east (14.451 km) 0, grid:south (24.535 km) 0, grid:north (29.962 km) 0, grid:west (35.93 km) 0
openweather	Surface pressure (pressure)	grid:center (10.991 km) 1013, grid:east (14.451 km) 1013, grid:south (24.535 km) 1013, grid:north (29.962 km) 1013, grid:west (35.93 km) 1013
openweather	Air temperature (temperature)	grid:center (10.991 km) 31.13, grid:east (14.451 km) 30.19, grid:south (24.535 km) 30.28, grid:north (29.962 km) 30.6, grid:west (35.93 km) 30.48
openweather	Wind direction (wind_direction)	grid:center (10.991 km) 229.7, grid:east (14.451 km) 215.2, grid:south (24.535 km) 217.6, grid:north (29.962 km) 199.4, grid:west (35.93 km) 199.3
openweather	Wind speed (wind_speed)	grid:center (10.991 km) 1.872, grid:east (14.451 km) 2.482, grid:south (24.535 km) 2.31, grid:north (29.962 km) 1.893, grid:west (35.93 km) 1.441
purpleair	PM2.5 (pm25)	99797 (0.6 km) 8.601, 166435 (2.677 km) 11.78, 97279 (2.684 km) 10.85, 6752 (5.276 km) 6.283, 222601 (6.699 km) 4.649, 133994 (7.377 km) 0.00274, 235425 (8.237 km) 8.504, 229853 (8.54 km) 4.805, +64 more
tceq	Carbon monoxide (co)	482011035 (0.0 km) 0.1533, 482011039 (14.355 km) 0.1098, 482011052 (15.44 km) 0.2609
tceq	Nitrogen dioxide (no2)	482011035 (0.0 km) 7.126, 482010416 (6.378 km) 5.121, 482011039 (14.355 km) 5.046, 482010026 (14.878 km) 6.746, 482011052 (15.44 km) 7.128, 482011015 (17.437 km) 8.775, 482010024 (19.743 km) 0.5348, 482011066 (22.737 km) 8.01, +4 more
tceq	Sulfur dioxide (so2)	482011035 (0.0 km) -0.2833, 482011039 (14.355 km) 0.3483, 482010051 (24.236 km) 0.2726


Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly
asos (9)
openaq (96)
purpleair (72)
tceq (13)
anomaly
50 km
40 km
30 km
20 km
10 km
Latitude (degrees north)
Longitude (degrees east)
30.1
30.0
29.9
29.8
29.7
29.6
29.5
29.4
−95.6
−95.4
−95.2
−95.0


Event-nearest wind vectors from stored values (arrow points downwind)
asos
longest arrow = 5.66 m/s
noaa_gfs
longest arrow = 4.42 m/s
openweather
longest arrow = 3.13 m/s

Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide
Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.
Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.
Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.
Missing or thin evidence means Unsure, not Invalid.
A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.
Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

A recirculated ship-channel or industrial-corridor plume is plausible because surface winds near the event time were variable but included southeasterly to easterly components at nearby observations and a southerly 10 m flow in GFS, which is consistent with onshore or sea-breeze-type transport into eastern Houston where sulfur-emitting industrial and marine sources are concentrated.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 2 of 36

A weather pattern or storm system is bringing in air from a nearby region with poor air quality, contributing to the anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 3 of 36

Atmospheric boundary layer dynamics during the afternoon facilitated the downward mixing of elevated industrial SO_2 stack plumes to ground level.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 4 of 36

The air quality anomaly in Houston, Texas on July 20th was caused by a combination of factors including high temperatures, low wind speeds, and nearby industrial activities.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 5 of 36

ASOS weather stations recorded southerly to southeasterly surface winds between 110 degrees and 176 degrees at speeds of 2.0 to 2.8 m/s near the time of the anomaly, consistent with plume transport from industrial facilities located south of the monitor.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 6 of 36

The PM2.5 levels were significantly higher than usual, with readings up to 149.3 ug/m3, indicating a high level of particulate matter in the air.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 7 of 36

A nearby short-duration industrial sulfur plume is supported because the TCEQ SO2 monitor at the anomaly site measured 1.8 ppb at 2026-07-20 22:00 UTC against an expected value of about -0.40 ppb, while the surrounding late-afternoon environment was hot, relatively dry, nearly cloud-free, and only weakly to moderately ventilated with surface winds near 2 to 3 m/s and a deep daytime boundary layer, conditions that favor transport of a concentrated local plume to a receptor without washout.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 8 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 9 of 36

The PM2.5 levels are elevated, with a mean of 9.0727 ug/m3 and a maximum value of 149.3 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 10 of 36

The temperature inversion over the Houston area is causing a buildup of pollutants near the surface.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 11 of 36

TCEQ surface monitors recorded a severe SO2 spike reaching 1.8 ppb at 22:00 UTC on July 20, 2026, which aligns with an increase in total SO2 column density observed by Sentinel-5P satellite granules over the Houston area.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 12 of 36

NO2 levels were also elevated, with readings up to 44.3 ppb, suggesting that nearby industrial activities or vehicle emissions may have contributed to the anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 13 of 36

The air quality anomaly in Houston, Texas is caused by high levels of NO2 particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 14 of 36

The nearest surface meteorology around 2026-07-20T22:00:00+00:00 showed hot and relatively dry conditions with temperature near 35.0 to 36.3 degC, humidity near 41.62 to 52.0 percent, low cloud cover near 6 percent, and wind speeds near 2.06 to 2.86 m/s.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 15 of 36

Localized emissions from industrial petrochemical and refining operations near the Houston Ship Channel caused a sharp increase in surface sulfur dioxide concentrations.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 16 of 36

A nearby industrial facility or power plant is releasing excessive amounts of particulate matter and other pollutants into the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 17 of 36

A monitor-scale artifact or a very localized plume that did not strongly affect the broader urban atmosphere remains a secondary explanation for the sulfur dioxide anomaly, because co-located carbon monoxide and nitrogen dioxide were not simultaneously elevated and the earlier Sentinel-5P sulfur dioxide column did not show a corresponding strong column enhancement near the site.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 18 of 36

GFS atmospheric model data indicates daytime planetary boundary layer heights peaking above 1200 meters during the afternoon of July 20, 2026, providing conditions for vertical convective mixing of elevated industrial emissions down to ground level.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 19 of 36

A short-duration sulfur dioxide emission from nearby industrial combustion or sulfur-handling sources in the Houston Ship Channel area is the leading physical explanation for the 1.8 ppb sulfur dioxide spike at the eastern Houston TCEQ monitor, because the anomaly was large relative to the monitor baseline while local winds were light to modest and variable enough to allow a narrow source plume to intersect the site.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 20 of 36

The TCEQ SO2 monitor at the anomaly location measured an unusually large positive sulfur dioxide excursion of 1.8 ppb at 2026-07-20T22:00:00+00:00 against an expected value of about -0.40 ppb, indicating a severe local SO2 event at that site.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 21 of 36

The tceq sulfur dioxide measurement at 29.734, -95.258 was 1.8 ppb at 2026-07-20T22:00:00+00:00, compared with an expected value of -0.402112676056338 ppb, yielding a z-score of 18.15417121179979 and a severe anomaly classification.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 22 of 36

A monitor-scale artifact or an extremely hyperlocal plume remains possible because the SO2 spike was not accompanied by clear concurrent increases in colocated TCEQ CO or NO2, nearby particulate measurements were not unusually elevated, and the nearest Sentinel-5P SO2 column snapshot occurred about two hours earlier and did not show a correspondingly strong column enhancement over the site.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 23 of 36

TCEQ ground-level SO2 measurements showed elevated 6-hour mean concentrations of 1.906 ppb during the 18:00 UTC period on July 20, 2026, compared to an overall context mean of 0.1143 ppb.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 24 of 36

Southerly to southwesterly surface winds transported sulfur dioxide plumes from upwind industrial facilities toward the monitoring station.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 25 of 36

The air quality anomaly in Houston, Texas is caused by high levels of SO_2 particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 26 of 36

TCEQ surface monitors recorded a severe SO_2 concentration anomaly of 1.8 ppb with a z-score of 18.15 at 2026-07-20T22:00:00+00:00.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 27 of 36

Meteorological conditions near eastern Houston at the anomaly time were hot, mostly clear, dry relative to the overnight period, and only lightly to moderately ventilated, which is consistent with transport of a narrow local plume and incomplete dilution.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 28 of 36

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 29 of 36

The tceq sulfur dioxide network within 50 km had a 6-hour mean of 1.906 ppb for 2026-07-20 18Z, which was much higher than the preceding 2026-07-20 12Z mean of -0.175 ppb and the following 2026-07-21 06Z mean of -0.2778 ppb.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 30 of 36

A transported and recirculated industrial plume from the southeast to south sector is a plausible physical explanation for the sulfur dioxide anomaly, because the anomaly occurred during hot, dry, nearly cloud-free late-afternoon conditions with a deep daytime boundary layer and low precipitation, a meteorological pattern consistent with sea-breeze-type redistribution and intermittent fumigation of elevated or distant ship-channel emissions.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 31 of 36

The anomaly occurred on July 20th, with the highest PM2.5 levels recorded at 21:59:59+00:00.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 32 of 36

Sentinel-5P satellite measurements recorded an increased 6-hour mean SO_2 column density of 1.73e-05 mol/m^2 during the 18:00 UTC timeframe on July 20, 2026.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 33 of 36

ASOS surface weather observations indicated southerly wind direction and warm daytime temperatures around 35 degrees Celsius at the time of the SO_2 peak.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 34 of 36

The absence of a strong coincident regional co-pollutant or column-scale sulfur signal supports a short-duration local industrial or ship-channel sulfur plume as the most likely cause rather than a broad regional air-quality episode.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 35 of 36

A severe sulfur dioxide (SO_2) air quality anomaly was detected by TCEQ ground monitors at location (29.734, -95.258) on July 20, 2026, at 22:00:00 UTC, reaching a value of 1.8 ppb with a z-score of 18.15.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 36 of 36

NOAA GFS model simulations showed a deep afternoon planetary boundary layer height reaching approximately 1200 meters near the anomaly location.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.